



Robust multi-lane detection and tracking using adaptive threshold and lane classification

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Abstract

Many global automotive companies have been putting efforts to reduce traffic accidents by developing advanced driver assistance system (ADAS) as well as autonomous vehicles. Lane detection is essential for both autonomous driving and ADAS because the vehicle must follow the lane. However, existing lane detection algorithms have been struggling in achieving robust performance under real-world road conditions where poor road markings, surrounding obstacles, and guardrails are present. Therefore, in this paper, we propose a multi-lane detection algorithm that is robust to the challenging road conditions. To solve the above problems, we introduce three key technologies. First, an adaptive threshold is applied to extract strong lane features from images with obstacles and barely visible lanes. Next, since erroneous lane features can be extracted, an improved RANdom SAMple Consensus algorithm is introduced by using the feedback from lane edge angles and the curvature of lane history to prevent false lane detection. Finally, the lane detection performance is greatly improved by selecting only the lanes that are verified through the lane classification algorithm. The proposed algorithm is evaluated on our dataset that captures challenging road conditions. The proposed method performs better than the state-of-the-art method, showing 3% higher True Positive Rate and 2% lower False Positive Rate performance.

Keywords Curved multi-lane detection and tracking · Feedback RANSAC · Lane classification · Local adaptive threshold · Kalman filter

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1 Introduction

1.1 Motivation

Currently, the global automotive industries and academia are actively researching autonomous driving and advanced driver assistance systems (ADAS). Perception technology is one of the most essential and important technologies to realize these systems because it can be a basis for the judgment that controls the vehicles. The perception technologies for autonomous vehicles and ADAS include lane detection, vehicle detection, road segmentation, sign recognition, and pedestrian detection. Among them, robust lane detection is essential for both autonomous driving and ADAS because false lane detection leads to wrong vehicle maneuvers.

For the lane recognition, computer vision-based perception system has been widely used to precisely detect lanes. In particular, a number of researchers focus on developing algorithms to detect multi-lane using computer vision, and these algorithms can benefit applications in Lane Departure Warning System (LDWS), Lane Keeping Assist System (LKAS),

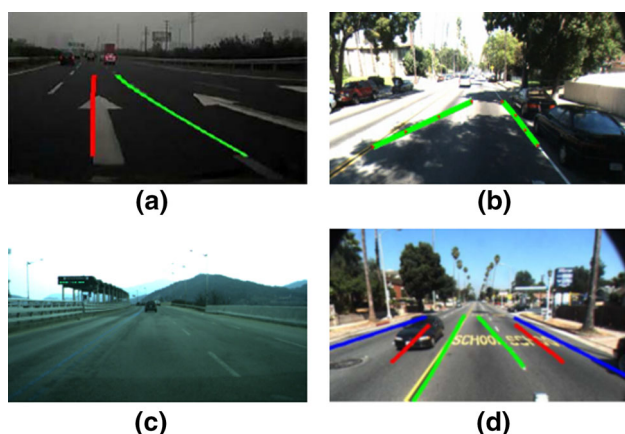


Fig. 1 Examples of false lane detection from challenging road conditions: **a** road marking [12], **b** shadow and light condition [2], **c** worn-out lane, **d** surrounding vehicle [2]

and localization of ego-vehicle. In reality, however, there are many difficulties in implementing vision-based multi-lane detection. Figure 1 describes various challenges associated with the lane detection. In general, vision-assisted lane detection technique aims to extract lane feature from an image. Therefore, road conditions with road markings in Fig. 1a and shadow in Fig. 1b cause false detection because the lane detection algorithms extract incorrect lane features interfered by such environments. The case of the worn-out lane in Fig. 1c is also challenging for lane detection due to weak lane feature. Furthermore, Fig. 1d shows that the presence of surrounding obstacles can lead to false lane detection. In this paper, we aim to solve these challenges by proposing a robust multi-lane detection algorithm where road markings, shadow, worn-out lanes, and surrounding obstacles are present.

1.2 Related works

In the literature, most studies try to overcome aforementioned challenges by proposing robust lane features. To strengthen the lane feature, the work of [8] integrates past frames with current frame to make the dashed lanes elongated. However, this work may result in false lane detection if the ego-vehicle performs lane change. Several approaches [19,28] propose symmetrical local threshold (SLT), which designates the lane features using the principle of the Dark-Light-Dark (DLD) transition property of lanes, to distinguish lanes from the noise by shadows. However, this approach takes an entire image frame as the input, so the presence of obstacles such as vehicles on the road would become noise, which results in false lane detection. Another work [17] introduces image enhancement of Inverse Perspective Mapping (IPM) image using a wavelet function to robustly detect a lane in a shadow condition, but this work focuses on lane detection in a shadow

condition and does not consider curved lane or the presence of obstacles. The other lane extraction method uses particle filter to strengthen the poor quality lane feature, and this method is found to detect worn-out lane [24]. However, this work utilizes costly radar and lidar, so lane detection system with affordable sensors would be preferred. These works focus on extracting robust lane features and may avoid false lane detection caused by road markings, shadows, and worn-out lanes, but there are challenges if the algorithms are to detect the lanes with the presence of obstacles on the road with affordable sensors.

Although selecting robust lane feature is crucial in multi-lane detection, the presence of surrounding obstacles such as vehicles and guardrail offers challenges because these obstacles occlude lanes or possess strong features that can interfere with lane feature extraction. In order to solve the above problem, many researchers [21,25,27] detect surrounding vehicles in advance for correcting the lane by assuming that the detected vehicles do not possess lane features. However, false vehicle detection results in false lane detection, so this approach requires robust vehicle detection. The other paper [9] utilizes the stereo camera to distinguish lanes on a road from obstacles based on disparity map, but the stereo vision-based system requires much process time to obtain three dimensions (3D) information.

Extracting robust lane features and filtering out surrounding obstacles are both required for lane detection that is robust to the challenging road conditions; however, detecting multi-lane is not trivial and should be considered as well. The background of multi-lane detection is surveyed in [18]. In the literature, the work of [13] uses the parallel characteristic of the multi-lane and predicts the possible locations for adjacent lanes using geometric lane estimation. This approach performs well if all the lanes are parallel; however, the proposed approach fails in case of merging lanes where non-parallel multi-lane exists. Another work [1] incorporates the information of traffic participants using GraphSLAM for multi-lane detection, but this work performs well only when some traffic participants are present. In our approach, we propose an adaptive threshold that is applied to smaller image regions so every lane is individually identified. Therefore, we do not assume previous knowledge of lane geometry or presence of traffic participants.

1.3 Contribution

The main contribution of the paper is that we propose a multi-lane detection algorithm that is robust to challenging road conditions (e.g., poor lane markings or occlusion by surrounding obstacles). To achieve this main contribution, we introduce three technologies an adaptive threshold, an improved RANSAC algorithm, and a lane classification algo-

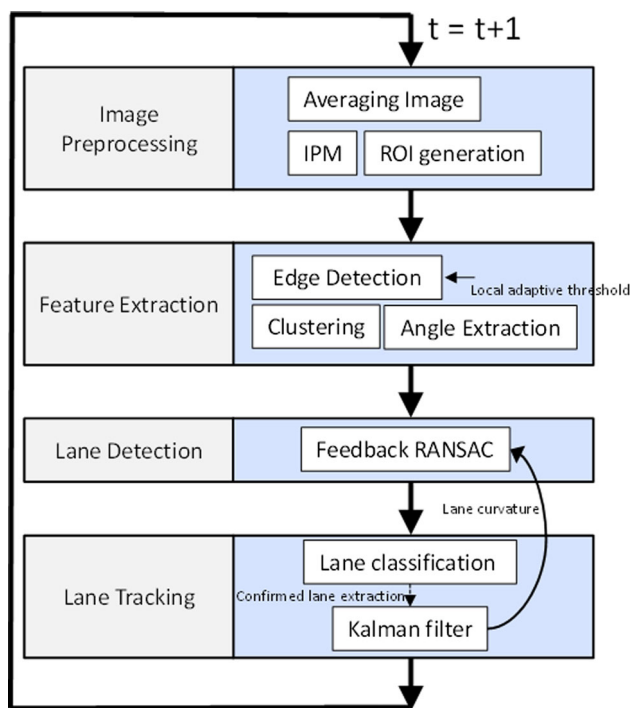


Fig. 2 Overview of the lane detection and tracking algorithm

algorithm and these technologies become a collective solution to the main contribution.

In order to enable multi-lane detection under challenging road conditions, we propose an adaptive threshold that is applied to smaller image regions. Each image region may include worn-out lanes or surrounding obstacles so we analyze the image pixel intensity of each region through the Gaussian Mixture Model (GMM) to identify strong lane features. Then, the multi-lane detection is performed by applying the Feedback RANSAC algorithm on these lane features, and we design two scoring functions to prevent the generation of false lanes due to surrounding obstacles. The lane classification algorithm filters out any noise and false lane in lane tracking. Overview of the proposed algorithm is illustrated in Fig. 2. As shown, the proposed robust multi-lane detection algorithm has four steps: image preprocessing, feature extraction, lane detection, and lane tracking.

The remainder of this paper is organized as follows. First, Sect. 2 introduces image preprocessing step. Obstacles such as skies, trees, and buildings are eliminated by specifying a region of interest (ROI). Next, inverse perspective mapping (IPM) is utilized to convert the input image into a top view image in order to eliminate the perspective effect. Then, the lanes are elongated by combining the current image with the past images. In Sect. 3, feature extraction is performed. The second derivative of the Gaussian filter is applied to the top view image to extract features of lanes, and an adaptive threshold is applied to extract only features of the correct

lane. Then, after clustering algorithm eliminates noise, a steerable filter is used to compute angles of the lane edge pixels. Lane detection algorithm is explained in Sect. 4. With a help of the edge angles and lane curvature history, Feedback RANSAC algorithm is newly proposed to improve the lane detection performance. Finally, Sect. 5 presents lane tracking technique, which completes the proposed lane detection and tracking algorithm. In this section, a detected lane is classified as either temporary lane or confirmed lane by lane classification algorithm based on the number of detected frames, and Kalman filter is applied for a smooth transition of lane tracking. In Sect. 6, the performance of the proposed algorithm is evaluated using various datasets. Concluding remarks and future works are provided in Sect. 7.

2 Image preprocessing

As the first step of the proposed algorithm, image preprocessing is performed. This section focuses on preprocessing the input image to provide appropriate features to be extracted in the feature extraction step.

2.1 Region of interest (ROI)

The role of the region of interest (ROI) extraction is to narrow the search region for lane detection. Many research studies have utilized ROI in calculating vanishing point or vanishing line [23,29,31]. Pre-assigning ROI to the image before the detection reduces the processing time and improves the accuracy of lane detection by eliminating irrelevant objects such as sky, buildings, and trees. In this work, the road is assumed to be flat, and ROI is selected as the region below the vanishing point, which is obtained by using projective model.

2.2 Inverse perspective mapping (IPM)

After the ROI generation, inverse perspective mapping transformation is applied on the sub-image. The IPM technique remaps the image pixels in camera view to a top view image. Many studies [2,4,6,10,14,17,26] utilize this technique to convert the input image into the top view image because it can remove the perspective effect and help the detection algorithm find the lane feature more easily. Once the IPM is applied, the cross-shape lane from the original image is changed into a parallel lane in the top view as shown in Fig. 3, and this parallel characteristic benefits the feature extraction stage [3,5]. In order to generate the top view image, each image pixel of the camera coordinates (u, v) are transformed into the top view image coordinates (x, y) , and the relation between (u, v) and (x, y) is formulated below with the parameters shown in Fig. 4:

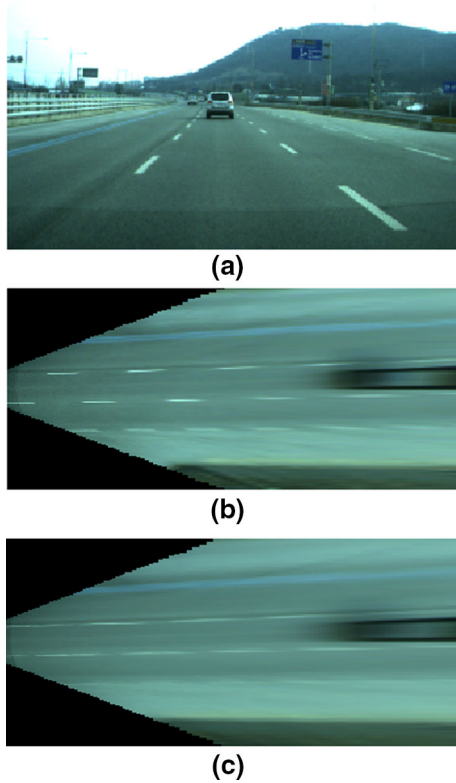


Fig. 3 Sequence of preprocessing algorithm. **a** Original image, **b** top view image, **c** temporal blurring image

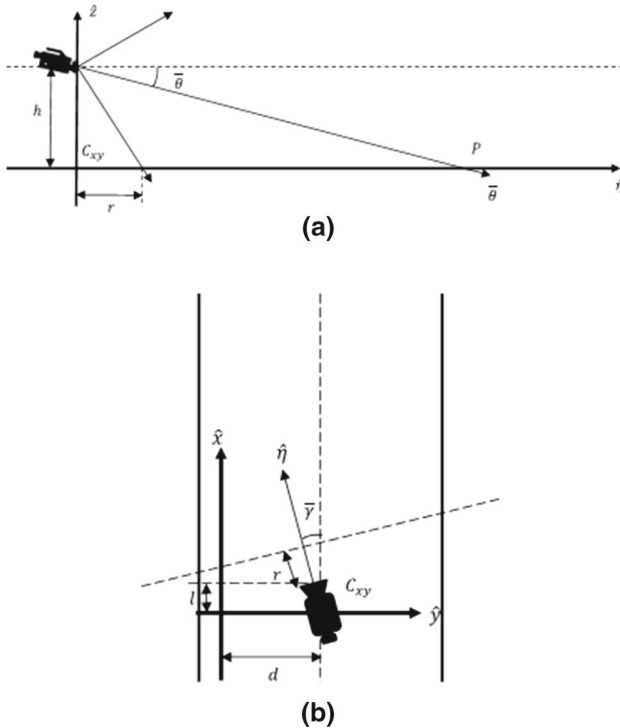


Fig. 4 Road model parameter illustration. **a** Side view, **b** road model parameter illustration

$$x(u, v) = h \times \cot[(\bar{\theta} - \alpha) + u(2\alpha/(m-1))] \times \cos[(\bar{\gamma} - \beta) + v(2\beta/(n-1))] + l \quad (1)$$

$$y(u, v) = h \times \cot[(\bar{\theta} - \alpha) + u(2\alpha/(m-1))] \times \sin[(\bar{\gamma} - \beta) + v(2\beta/(n-1))] + d \quad (2)$$

These equations are based on the assumption that the road is flat and can be derived by using extrinsic parameters and intrinsic parameters of the camera. Inversely, the camera image coordinates can be obtained from the top view coordinates, as described in the following:

$$u(x, y) = \frac{\cot \left[h \times \sin \left\{ \cot \left(\frac{y-d}{x-l} \right) \right\} / (y-d) \right] - (\bar{\theta} - \alpha)}{\frac{2\alpha}{m} - 1} \quad (3)$$

$$v(x, y) = \frac{\cot \left(\frac{y-d}{x-l} \right) - (\bar{\gamma} - \beta)}{\frac{2\beta}{n} - 1}, \quad (4)$$

The descriptions of extrinsic and intrinsic parameters of the camera are shown below:

- Field of view (aperture) is $2\alpha \times 2\beta$ radians,
- Resolution of the camera is $m \times n$ pixel,
- Yaw angle is $\bar{\gamma}$ radians,
- Pitch angle is $\bar{\theta}$ radians.

2.3 Lane extension via image superposition

When there appears worn-out or dashed lane, it is difficult to detect the lane from a single frame. By averaging pixel values of the lanes from two or three consecutive frames, the low-quality lanes can be improved to a lane that can be more easily detected [7,8]. Commonly, the method of integrating frames is called temporal blurring, and the resulted average image is represented as

$$\text{Averaged Image} = \sum_{i=0}^N \frac{I(n-i)}{N} \quad (5)$$

where i is the index offset of the frame and $I(n)$ is the current frame, and N is the number of frames used for the integration. N must be defined as the number of frames that are to be processed in a short period of time. From the experiments, the frames within 0.1 second are suitable to generate an averaged image, so N is selected as either two or three based on frame speed. The result of temporal blurring is shown in Fig. 3c.

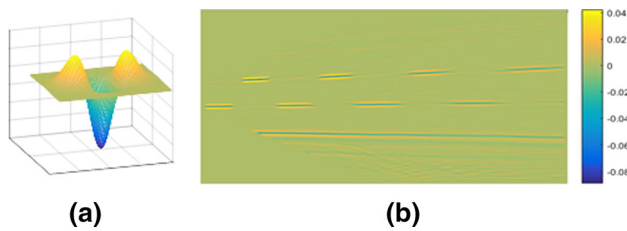


Fig. 5 **a** Second derivative Gaussian filter, **b** filtered image

3 Feature extraction

The stage of lane feature extraction is the core of lane detection algorithm because the performance of lane detection highly depends on the quality of the lane features. In particular, the road condition with the presence of surrounding obstacles and guardrail is a challenging scenario for feature extraction because the features of lanes are not always stronger than those of obstacles. Especially, the lane can be worn as well as the lane color may vary in the course of vehicle driving, which are the possible causes for weak lane features. Therefore, this paper proposes a local adaptive threshold to extract not only a clear lane but also an unclear lane. In addition, this proposed adaptive threshold method eliminates noises coming from surrounding vehicles and guardrail.

3.1 Lane edge extraction using adaptive threshold

First, a Gaussian filter is applied to the input image to smooth the image features [2,10,15,30]. Since the camera image is transformed to the top view image using IPM, the Gaussian filter works well in extracting the feature of the lane edges, and any other similar filter such as Gabor filter or Top-hat filter can be used to extract such feature showing similar performance. Here, the smooth image $L(x, y)$ is created by convoluting Gaussian filter $G(x, y)$ with the input image $I(x, y)$, as shown in the following:

$$\begin{aligned} L(x, y) &= I(x, y) * G(x, y) \\ &= I(x, y) * \frac{1}{2\pi\sigma^2} \exp(-(x^2 + y^2)/2\sigma^2) \end{aligned} \quad (6)$$

where x and y represent the horizontal and vertical position on the image, respectively. As the following step, the smoothed image is filtered to extract the edge features. In general, the lanes on the top view image are parallel and elongated horizontally, so taking derivatives in y -direction provides appropriate filters [2]. Thus, the second derivative of Gaussian filter $G_{yy}(x, y)$, which is shown in Fig. 5a, is applied to the smoothed image $L(x, y)$ to generate filtered image $B(x, y)$ shown in Fig. 5b, as described in the following:

$$\begin{aligned} G_{yy}(x, y) &= \frac{\partial^2}{\partial y^2} \exp\left(-\frac{x^2 + y^2}{\sigma^2}\right) \\ &= -\left(\frac{2y}{\sigma^2} - 1\right) \frac{2}{\sigma^2} \exp\left(-\frac{x^2 + y^2}{\sigma^2}\right) \end{aligned} \quad (7)$$

$$B(x, y) = L(x, y) * G_{yy}(x, y) \quad (8)$$

The filtered image $B(x, y)$ shows that the pixels for white lanes have a high negative intensity value. The clearer the white lane is, the more negative value the pixel of the filtered image becomes. In this case, it is possible to extract the clear white lane by applying low negative threshold value. However, in case of unclear lanes such as blue or worn-out lanes, the pixel intensity has a value close to zero. Therefore, in order to extract the unclear lane, a threshold close to zero should be applied. However, since the pixels for surrounding obstacles and guardrail are also found to have values close to zero, the feature points for the obstacles are likely to be extracted together with the dim lane, causing the false lane detection. Thus, in order to detect lanes including worn-out lanes and exclude the noises, an adaptive threshold is necessary for robust lane detection.

3.1.1 Analysis of road component

In order to systematically propose an adaptive threshold method, the filtered image pixels are analyzed through the Gaussian Mixture Model (GMM) by using the Expectation Maximization (EM) algorithm. Through GMM, the components of the road can be divided into the clear lane, unclear, and non-lane components as shown in Fig. 6. The clear lane component selects pixels of low negative intensity from the filtered image and present white lane as the output, depicted in Fig. 6b. Although the clear lane component can be used for white lane detection, the blue lane on the top of Fig. 6a belongs to the unclear component because the intensity value of the blue lane or dim lane is close to zero. The unclear component is found to present the presence of surrounding obstacles and dim lane, which have pixel intensities near zero. The non-lane component provides the remaining features of the filtered image, and their pixel intensities are above zero. By analyzing each component in detail, the GMM method presents that the pixel intensity from the filtered image increases in the order of 1. Clear lane 2. Obstacle 3. Dim lane 4. Non-lane. If a filtered image includes all these four components, it would be very challenging to apply the adaptive threshold technique; therefore, it is necessary to divide the filtered image into smaller image regions so that each region includes only a few of these four components. This work assumes that the length of a general dashed lane and the width between two lanes are already known, so the length and the width of the region is slightly smaller than those. As a result, each region cannot include clear and

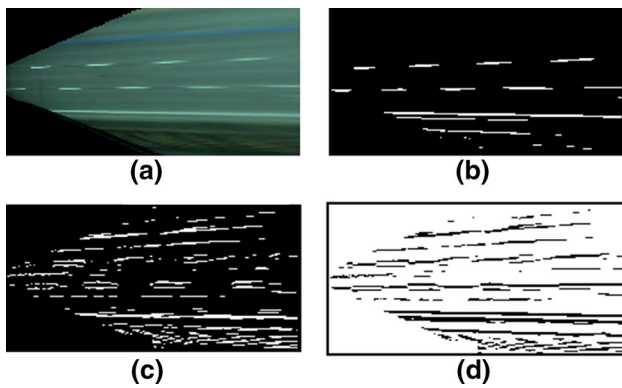


Fig. 6 Images corresponding to each component obtained by GMM. **a** Top view image, **b** clear lane component, **c** unclear component, **d** non-lane component

Table 1 The image region classification

The image region state	PBT _{ratio} state
Case 1: Dim lane + Non-lane	$PBT_{ratio} < \varepsilon_1$
Case 2: Clear lane + Non-lane	$\varepsilon_1 < PBT_{ratio} < \varepsilon_2$
Case 3: Obstacle + Dim lane + Non-lane	$\varepsilon_1 < PBT_{ratio}$
Case 4: Dim lane + Clear lane + Non-lane	$\varepsilon_2 < PBT_{ratio}$

dim lane simultaneously, and then the adaptive threshold is applied based on the state of each image region. Table 1 describes four possible states for each image region. Each region consists of a lane and non-lane with the possible presence of an obstacle. To distinguish different cases, PBT_{ratio} is defined as

$$PBT_{ratio} = \frac{\# \text{ of Pixels of value Below } Thresh_{class}}{\text{total \# of pixels from an image region}} \quad (9)$$

where Thresh_{class} is classification threshold value for distinguishing clear lane and obstacle from dim lane and non-lane. Since the ratio of pixels within Thresh_{class} increases as more portion of clean lane and obstacle is included in an image region, the four region cases show different ranges of PBT_{ratio}, limited by parameters ε_1 and ε_2 as shown in Table 1. In this paper, $\varepsilon_1 = 1/100$ and $\varepsilon_2 = 1/10$ are experimentally selected. Once the image region state is decided, the adaptive threshold is to be applied to each case in order to only extract the lane from the filtered image.

3.1.2 Assignment of adaptive threshold

Figure 7 illustrates the flowchart of how the adaptive threshold is assigned to different image region state. Initially, the filtered image from the previous step is divided into the smaller regions, and PBT_{ratio} is computed for each region. As the first scenario, if the calculated PBT_{ratio} is less than ε_1 , the state falls into case 1, which consists of dim lane and

non-lane. In this case, a preassigned threshold is applied to select the dim lane and exclude the non-lane. As the next scenario, if PBT_{ratio} is larger than ε_1 and smaller than ε_2 , the state falls into either cases 2 or 3 since the PBT_{ratio} state of case 3 does not have an upper bound. This ambiguity is due to the similar pixel intensity for obstacle and dim lane, and the noise caused by the obstacle is to be reduced by a clustering algorithm in the following section. In this scenario, the clear lane is detected by selecting the pixels with intensity of the lowest $\beta\%$. Since clear lane has the lowest intensity among the four road components, selecting the lowest $\beta\%$ is known to work well in clear lane detection. For the third scenario, if PBT_{ratio} is larger than ε_2 , the state falls into either cases 3 or 4. Similarly, the clear lane is detected by selecting the pixels with intensity of the lowest $\alpha\%$. As can be seen in the second and the third scenarios, the dim lane should be detected by removing the noise from obstacles, which will be explained in the following section. From the experiments, α and β are selected as 1 and 2.5, respectively. The output from this step is a binary image with the value of 1 for the selected pixels by adaptive threshold and value of 0 otherwise.

3.1.3 Noise reduction by clustering

Since the binary image from the previous step includes noises from surrounding vehicles and guardrail, noise reduction is performed by using a clustering algorithm. The main idea of the algorithm is that the pixels for a correct lane are close to each other and in the similar color. If there exists such group of pixels aligned together with the number of the pixels above the threshold, the group of pixels is clustered as a lane.

In Algorithm 1, the binary image pixels x and the pixel values in gray scale g from the filtered image are used as the initial inputs. N_x and N_c represent the number of all nonzero pixels in the binary image and the number of clustered pixels, respectively. For each binary image pixel, the distance and gray scale intensity are compared with every clustered pixel, and those within preassigned thresholds ϕ_1 and ϕ_2 are clustered to a group of pixels, where c denotes the group number of clustering. The function $\text{dist}(x_i, x_j)$ calculates the distance between the pixel x_i and x_j , and the function $|g(x_i) - g(x_j)|$ computes the gray scale values difference between the two pixels. If x_i is not affiliated with any existing group, a new group with the group number $\text{MAX}(c) + 1$ is created to include x_i . Finally, if the number of elements, denoted by $n(c_k)$, for a group is less than threshold ϕ_3 , the pixels x that belong to the group are excluded from the binary image. This logic removes the noise coming from the clustering groups.

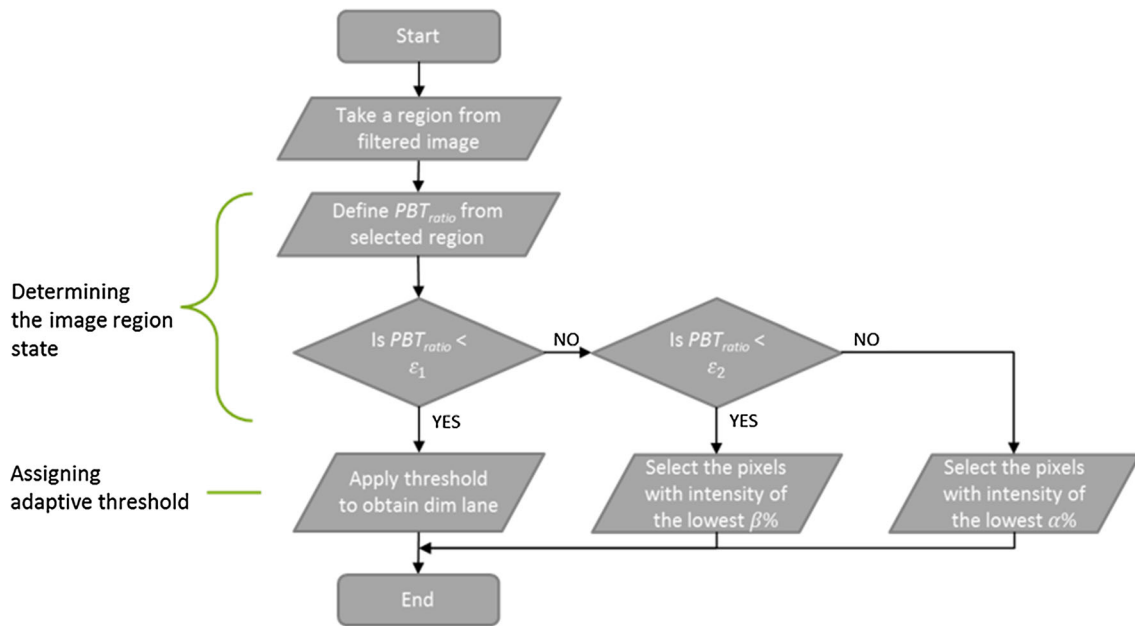


Fig. 7 Flowchart of applying adaptive threshold

Algorithm 1. Clustering Algorithm

Input: The binary image pixels (x) and the pixel values in gray scale (g)

```

1: For:  $i \leftarrow 1$  to  $N_x$ 
2: For:  $j \leftarrow 1$  to  $N_c$ 
3: If  $dist(x_i, x_j) < \phi_1$  and  $|g(x_i) - g(x_j)| < \phi_2$ 
4:    $c(x_j) \leftarrow c(x_i)$  and  $N_c + 1 \leftarrow N_c$ 
5: end If
6: end For
7: If  $x_i$  is not included in any  $c$ 
8:    $MAX(c) + 1 \leftarrow c(x_i)$  and  $N_c + 1 \leftarrow N_c$ 
9: end If
10: end For
11: For  $k \leftarrow 1$  to  $MAX(c)$ 
12: If  $n(c_k) < \phi_3$ 
13:    $0 \leftarrow x$  for  $x \in c_k$ 
14: end If
15: end For

```

Output: The clustered binary image

Figure 8 shows the comparison of existing threshold application method with the proposed adaptive threshold method in the presence of obstacles. As can be seen in Fig. 8b, when a general threshold is applied, the blue dim lanes are not detected because the pixel intensity of the filtered image for the dim lane is below the threshold. On the contrary, when a threshold close to zero is applied to the binary image in order to detect the dim lanes, as shown in Fig. 8c, many noises are generated by obstacles. Figure 8d shows the binary images after applying the proposed adaptive threshold, which removes noise by classifying clear lanes and dim

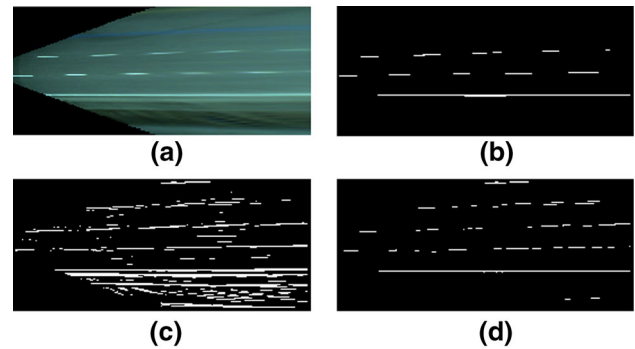


Fig. 8 The comparison of binary images with different thresholds when there are noises caused by guardrail. **a** Top view image, **b** the binary image after applying a general threshold, **c** the binary image after applying threshold close to zero, **d** the binary image after applying the proposed adaptive threshold

lanes from non-lanes. The noises caused by obstacles are further reduced by clustering, as shown in Fig. 9, which illustrates the overview of lane feature extraction.

3.2 Edge angle extraction

This section covers angle extraction (Fig. 9d) from edge features (Fig. 9c) obtained from the previous step. An angle of the lane feature is very helpful in lane detection [16,22,26] because the angle can be used as the scoring function in later RANSAC algorithm. In this paper, a steerable filter [11] combined with the second derivative of Gaussian is utilized to measure angles corresponding to the features, as shown in the following:

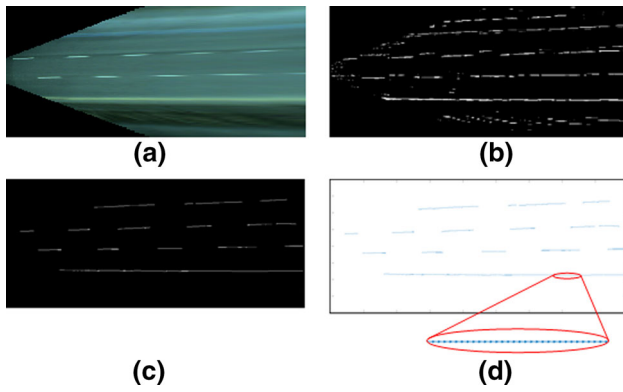


Fig. 9 Sequence of feature extraction. **a** Top view image, **b** the binary image before clustering algorithm, **c** the binary image after clustering algorithm, **d** the angles of edge features

$$G_{xx}(x, y) = \frac{\partial^2}{\partial x^2} \exp\left(-\frac{x^2 + y^2}{\sigma^2}\right) = -\left(\frac{2x}{\sigma^2} - 1\right) \frac{2}{\sigma^2} \exp\left(-\frac{x^2 + y^2}{\sigma^2}\right) \quad (10)$$

$$G_{yy}(x, y) = \frac{\partial^2}{\partial y^2} \exp\left(-\frac{x^2 + y^2}{\sigma^2}\right) = -\left(\frac{2y}{\sigma^2} - 1\right) \frac{2}{\sigma^2} \exp\left(-\frac{x^2 + y^2}{\sigma^2}\right) \quad (11)$$

$$G_{xy}(x, y) = \frac{\partial^2}{\partial x \partial y} \exp\left(-\frac{x^2 + y^2}{\sigma^2}\right) = \frac{4xy}{\sigma^4} \exp\left(-\frac{x^2 + y^2}{\sigma^2}\right) \quad (12)$$

$$G2^\theta(x, y) = G_{xx} \cos^2 \theta + G_{yy} \sin^2 \theta + G_{xy} \cos \theta \sin \theta \quad (13)$$

where θ represents an angle which rotates on G_{xx} axis. By differentiating (13) and setting the derivative to zero, the minimum and the maximum angles are calculated. The angles can be obtained as

$$\theta_{\min} = \arctan\left(\frac{G_{xx} - G_{yy} + A}{2G_{xy}}\right) \quad (14)$$

$$\theta_{\max} = \arctan\left(\frac{G_{xx} - G_{yy} - A}{2G_{xy}}\right) \quad (15)$$

where

$$A = \sqrt{G_{xx}^2 - 2G_{xx}G_{yy} + G_{yy}^2 + 4G_{xy}^2} \quad (16)$$

The minimum angle is corresponding to the slope of lanes, as shown in Fig. 9d. The minimum angles of the edge features from edge angle extraction become the final output of feature extraction stage for lane detection.

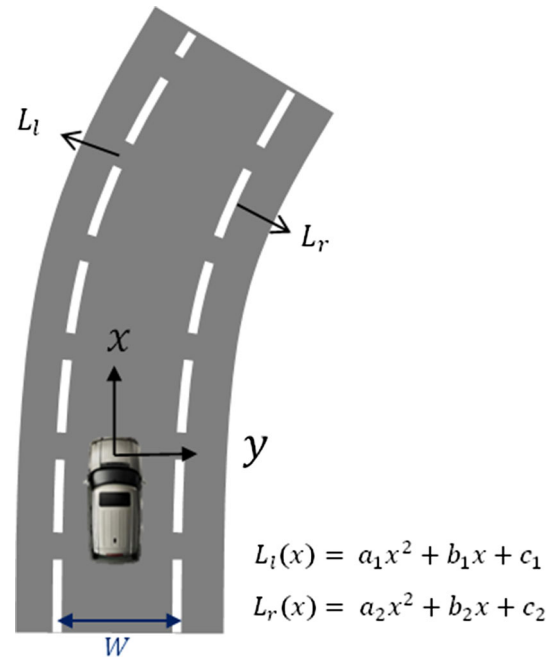


Fig. 10 Illustration of road and lane model on the world coordinate

4 Lane detection

4.1 Lane model

The lane features obtained from the previous section are used by a lane model to detect the lane, so determining proper lane model is critical. In the literature, researchers use lane models that are straight, spline, and polynomial model. These models are applied in accordance with various kind of driving environments on a highway or urban. In highway, the curvature of the road is constrained because vehicles drive at high speed. Therefore, a quadratic model [20,26] is suitable for representing the highway lanes, as shown in Fig. 10, and utilized in this work. The quadratic model can calculate lateral offset, direction angle of the vehicle, and curvature of the road. The model formula is

$$y = ax^2 + bx + c \quad (17)$$

where x and y are the coordinates shown in Fig. 10 and a , b and c are the parameters of the quadratic model, which are used to calculate lateral deviation Δy , the slope of lane μ , and curvature of road κ , as shown in the following:

$$\Delta y = -c \quad (18)$$

$$\mu = -\arctan(b) \quad (19)$$

$$\kappa = 2a/(b^2 + 1)^{3/2} \quad (20)$$

4.2 Feedback RANSAC

Algorithm 2. Feedback RANSAC algorithm

- 1) Select a small number of pixels from the lane edge features
- 2) Fit a quadratic model $y = ax^2 + bx + c$ using least mean square on the selected pixels.
- 3) Compute the scoring function for the fitted model.
- 4) Repeat the steps from 1) to 3) for n times and select the model with the highest score.
- 5) If the highest score is above threshold, save the selected lane model. Otherwise, return to the step 1).
- 6) Once a model is saved, select pixels that are passed by the model, and refit those to a new quadratic lane model.
- 7) Repeat the steps from 1) to 6) for m times to extract all lanes including ego and adjacent lanes.

In this part, the outliers of lane edge features are removed an improved RANSAC algorithm and the filtered lane features are used to fit a quadratic lane model by choosing the parameters a , b , and c . Although there are various lane detection techniques, they face challenges when dealing with a wide range of lane conditions. For instance, Hough transform is only useful for extracting the straight lane, not the curved lane, and least mean square method can be easily influenced by outliers caused by obstacles. Thus, Algorithm 2 proposes to use an improved Random Sample Consensus (RANSAC) algorithm, namely Feedback RANSAC, which use the feedback from the lane edge angles and the lane curvature history. RANSAC-based lane detection is a technique of eliminating outliers by repeatedly fitting lanes and then selecting parameters of the model with the highest score [2,6,10,12]. The Feedback RANSAC algorithm randomly selects pixels and then calculates the score of RANSAC on the fitted quadratic model. Although RANSAC generally uses a single scoring function, the Feedback RANSAC uses two scoring functions shown below to solve aforementioned challenges.

$$\text{score}_1 = \frac{\# \text{ of nearby pixels with similar edge angle}}{\text{Maximum possible \# of selected pixels}} \quad (21)$$

$$\text{score}_2 = \tanh\left(\frac{1}{20 * |\kappa_i - \kappa_{i-1}|}\right) \quad (22)$$

$$\text{score} = \alpha * \text{score}_1 + (1 - \alpha) * \text{score}_2 \quad (23)$$

where α is the weight of scoring, i is the frame number, and κ is the lane curvature from (20). The parameter α is experimentally chosen as 0.5 for the optimal performance.

Equation (21) describes the first scoring function, which counts the number of pixels with the edge angle similar to the slope of the selected lane and also within a threshold distance, as shown in Fig. 11a. The first score represents how precisely the extracted lane matches with the extracted edge features within a desired threshold distance. However, this function still includes outliers caused by road markings, surrounding vehicles, and guardrails. Therefore, a second scoring function is utilized to solve this problem. The second

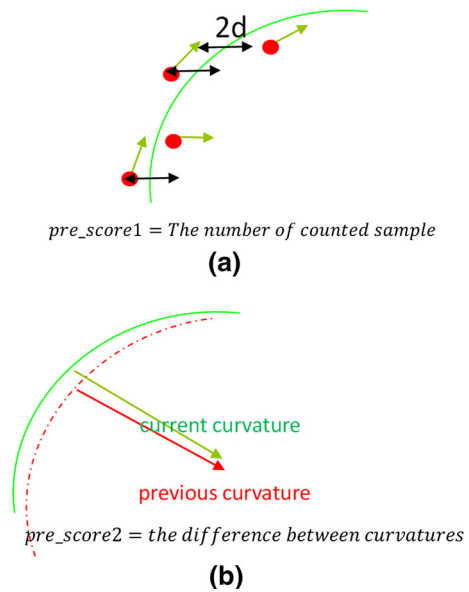


Fig. 11 Two scoring functions of the Feedback RANSAC algorithm. **a** First scoring, **b** second scoring

scoring function calculates the difference between current and previous curvature as shown in Fig. 11b and gives a low score for the outliers with large deviation caused by obstacles. Equation (22) shows that the second scoring function uses hyper-tangent function so that the weight of the score is comparable to the first scoring function.

As the next step, if the fitted model with the highest score has a score above the desired threshold, the lane model is saved and refitted to a new quadratic model once again using the selected pixels. The procedure for obtaining a refined lane model is repeated to detect all lanes including ego- and adjacent lanes.

5 Lane tracking

In this section, lane classification method and Kalman filter technique are explained to perform robust lane tracking. As Fig. 12 illustrates, the overall framework for lane tracking, the detected lanes from the previous section are first classified into *Confirmed Lane*, *Temp Lane*, or *Extinct Lane* via lane classification, and only the Confirmed Lane is tracked by Kalman filter through measurement and time update.

5.1 Lane classification

Before performing tracking algorithm, it is necessary to associate the detected lanes from the current frame with the lanes from the previous frame because the lanes are extracted in random order. The lane association is determined by computing the lateral distance between the lanes and selecting the

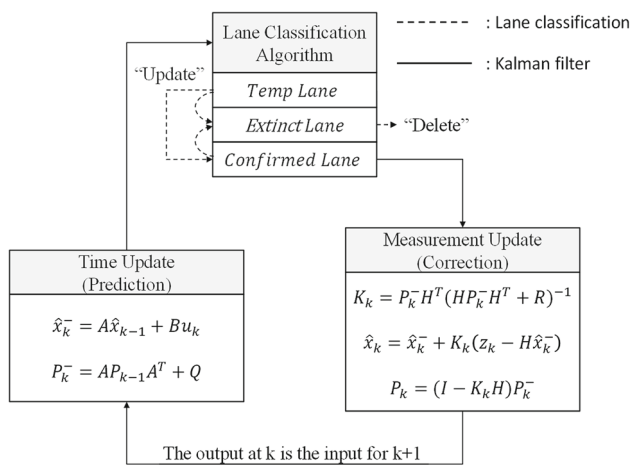


Fig. 12 Overall framework for lane tracking with lane classification

Table 2 Lane classification

Confirmed lane	Temp lane	Extinct lane
An associated lane is detected more than n times	An associated lane is detected less than n times	A confirmed lane is undetected more than m times

ones below a threshold. In each frame, the associated lane is classified into one of three possible lanes by the criteria described in Table 2.

Table 2 shows the lane classification method, which defines three types of lanes *Confirmed Lane*, *Temp Lane*, and *Extinct Lane*. The associated lane is classified as *Confirmed Lane* when it is detected more than n times, and as *Temp Lane* when it is detected less than n times. The *Temp Lane* is considered a false lane and not used in Kalman filter. Moreover, if a *Confirmed Lane* is undetected more than m times, it becomes *Extinct Lane*, which is to be deleted from the queue. From the experiments, m and n are decided as the number of frames within 0.2 s.

5.2 Kalman filter

As for the final step of lane tracking, Kalman filter technique is implemented to track lanes [6,10,26,29]. Kalman filter is suitable for our application since the lane model is fixed as the quadratic so it gives a nice Gaussian solution. Other filters such as particle filter may be also considered but multi-lane detection requires a number of particles to be tracked throughout the image so Kalman filter is simple and preferred in this work. From the lane classification, *Confirmed Lane* is used for tracking with the Kalman filter model defined as

$$x(k+1) = Ax(k) + Bu(k) + w(k) \quad (24)$$

$$z(k) = Hx(k) + v(k) \quad (25)$$

where $x(k)$ is the state variable, $z(k)$ is the state of the measurement, $w(k)$ and $v(k)$ are random variables, and $u(k)$ is the control input. In lane tracking, there is no control input, so we can neglect the term with $u(k)$, and matrices A and H are part of the system model. The detailed expressions are as follows:

$$x(k) = [a \quad \dot{a} \quad b \quad \dot{b} \quad c \quad \dot{c}]^T \quad (26)$$

$$z(k) = [a \quad b \quad c]^T \quad (27)$$

$$w(k) \sim N(0, Q), \quad v(k) \sim N(0, R) \quad (28)$$

$$Q = \begin{bmatrix} \sigma_1 & 0 & 0 & 0 & 0 & 0 \\ 0 & \sigma_2 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma_1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sigma_1 & 0 \\ 0 & 0 & 0 & 0 & 0 & \sigma_2 \end{bmatrix},$$

$$R = \begin{bmatrix} \sigma_3 & 0 & 0 \\ 0 & \sigma_3 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & dt & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & dt & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & dt \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \quad (29)$$

Here, the state variable $x(k)$ consists of the parameters of the quadratic lane model and the rate of change of the parameters. The rate of change is defined as the difference of the parameter between the lanes of the current frame and those of the previous frame. The state of the measurement $z(k)$ consists of the parameters of the quadratic lane model. The random variables $w(k)$ and $v(k)$ represent the process and the measurement noise, respectively. Covariance Q and R have white Gaussian probability distribution with zero average. The covariance values are used for smooth lane transition in lane tracking, and the parameters σ_1 , σ_2 , and σ_3 are experimentally set as 1, 3, and 50. The matrices A and H are the system models, and dt represents the time interval between consecutive frames.

6 Experimental result

In order to verify the proposed system, the lane detection and tracking algorithm is tested with two datasets, which are

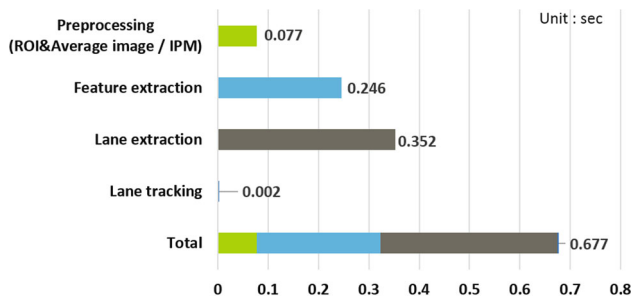


Fig. 13 The process time in each step

comprised of various road environments. The first dataset consists of scenes from the roads near KAIST in Daejeon, South Korea. For collecting this dataset, VSTC-V200G camera was mounted on a vehicle. The test video is composed of 4335 frames with a resolution of 640 by 360 pixels, and the speed of the image input is 20 frames per second. The lane detection performance with this dataset is introduced in Sect. 7.1. The second dataset, Caltech Lane Datasets [1], is used to compare the performance of our algorithm with other state-of-the-art algorithms, which will be addressed in Sect. 7.2. The proposed algorithm is tested on a general desktop with Intel(R) Core(TM) i7-4770 CPU at 3.40 GHz and 8.00 GB RAM. Figure 13 shows that the total process time for applying the algorithm on an image frame with the MATLAB R2015b takes 0.677 s, and the process time for an individual step is also shown. As can be found from Fig. 13, lane extraction step takes the most time because the Feedback RANSAC algorithm repeats the iterations for searching the optimal lane with the highest score. As shown in Fig. 14, split lane, curved lane, multi-lane with road markings or arrows, and worn-out lane are correctly detected as well as lanes with the presence of surrounding vehicles and guardrail.

Table 3 Performance evaluation and analysis of the dataset

Evaluation and analysis	# of lanes
Total frames	4335
Total lanes	8670
True positive (TP)	8579
False positive (FP)	8
True negative (TN)	—
False negative (FN)	91
Precision	99.9%
Recall	99.0%
F-measure	99.4%
Accuracy	98.9%

6.1 Lane detection performance

In this section, the lane detection performance is evaluated using the first dataset. The collected dataset consists of adverse road conditions, and Table 3 provides the details of the tested dataset and results. As Fig. 14a, e demonstrates, the algorithm successfully detects lanes when obstacles are present on the road, and Fig. 14h shows that the lane detection works well even if the lanes are worn-out as can be seen in Fig. 14g. In evaluating the performance, we focus on finding False Positive for lane detection in all these adverse conditions, and our supplemental video qualitatively confirms that our algorithm is a robust solution to the challenging conditions. The ground truth is obtained by manually checking each frame, and a detected lane is considered true if it lies on the lane marking. It is challenging to generate ground truth for adjacent lanes due to occlusion by obstacles, so this paper focuses on evaluating the result based on ego-lane detection.

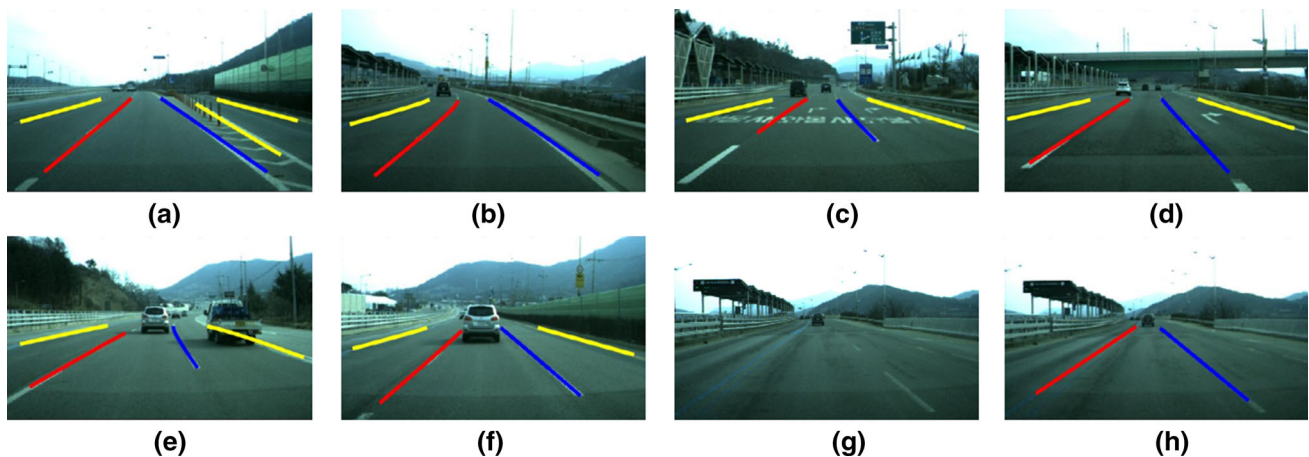


Fig. 14 The results of the proposed algorithm on various road environments. **a** Split lane detection with separators, **b** curved lane with guardrail, **c** multi-lane with road markings, **d** multi-lane with road arrow, **e** multi-lane with surrounding vehicles, **f** multi-lane with close-in surrounding vehicles, **g** the original image of worn-out lane, **h** the result of worn-out lane detection

e multi-lane with surrounding vehicles, **f** multi-lane with close-in surrounding vehicles, **g** the original image of worn-out lane, **h** the result of worn-out lane detection

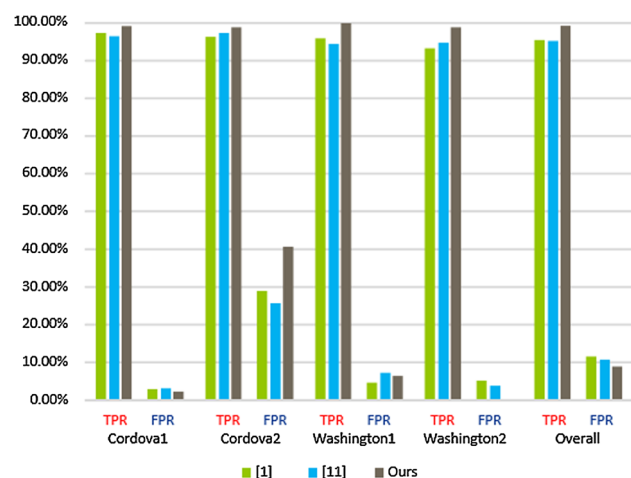


Fig. 15 Performance comparison with the work of [2] and [12]

For the evaluation, a total of 8670 lanes from 4335 frames are considered, which results in True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN). The value of TN is unknown since lane detection does not predict false lane for the false lane. Among a total of 8670 ego-lanes of the dataset, the number of TP is 8579, the number of FN is 91, and the number of FP is 8. These values lead to a very high precision of 0.999, recall of 0.990, F-measure of 0.994, and accuracy of 0.989. The precision indicates if the extracted lanes accord with the actual lanes so high precision prevents the vehicle from deviating from its usual route. High recall means that almost all the actual lanes are detected, which contributes to a good lane keeping system. F-measure is used to see the overall performance by taking the harmonic average of the precision and recall. These results demonstrate that the rate of false lane detection is very low and almost all ego-lanes are recognized. In case of the worn-out or blue lane, the proposed adaptive threshold helps the detection algorithm extract the strong features of the lanes. In other cases when road markings and surrounding obstacles are present, false lanes are prevented by feedback RANSAC and lane classification.

6.2 Comparison with the state-of-the-art algorithms

To verify the performance of the proposed algorithm, the proposed work is compared with the work of [2] and [12] using Caltech Lane Datasets [2]. The Caltech Lane Datasets consist of four types of urban driving scenarios: Cordova1, Cordova2, Washington1, and Washington2, with a total of 1224 frames containing 4172 lane markings. For each driving scenario, the True Positive Rate (TPR) and False Positive Rate (FPR) are calculated and compared in Fig. 15.

The results show that the performance of our algorithm is superior to other existing performances. The averages of TPR

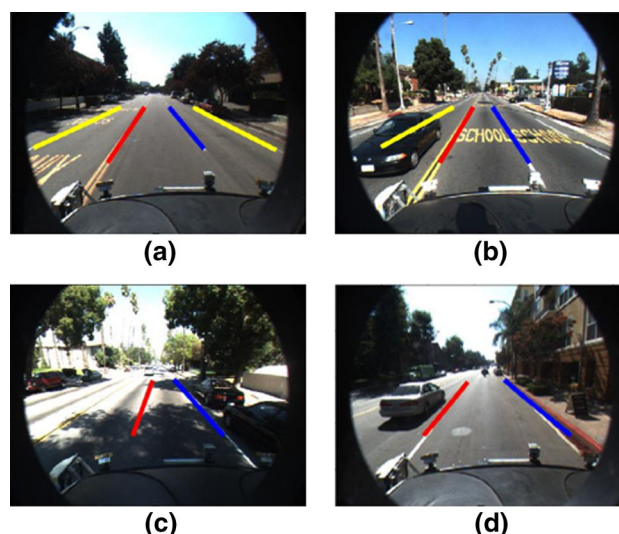


Fig. 16 The result of our algorithm on the Caltech Lane Datasets. **a** Road marking, **b** road marking and obstacle, **c** shadow and light, **d** surrounding vehicle

and FPR for our work are 98.99% and 8.88%, respectively, which demonstrates the high detection rate of our algorithm. Especially, the dataset of the Washington1 video includes many shadow and light environments, but the performance of our algorithm outweighs the performances of other two works. In case of the Cordova2 dataset, our algorithm shows the higher FPR because our algorithm perceives the curb of the sidewalk as a lane. Overall, the result concludes that our algorithm is able to detect the lanes from various challenging scenarios and thus minimize the detection error to be less than two percent, which would not be possible with conventional algorithms.

Figure 16 shows the result of the proposed algorithm on the Caltech Lane Datasets. It is challenging to detect lanes in these various environments because false lanes can be generated by following the shape of shadow or obstacle as previously shown in Fig. 16b, d. It can be seen that our algorithm is robust to road markings, shadow, illumination change, and the presence of surrounding obstacles because false lanes are prevented by the proposed adaptive threshold and lane classification.

7 Conclusion and future works

In the paper, the robust multi-lane detection and tracking algorithm is proposed for autonomous driving in various environments on the highway. Previous research has limitations in detecting worn-out lanes, blue lanes, and lanes under illumination changes or shadow because their lane features are difficult to be extracted. In order to solve the problems, this paper introduces the local adaptive threshold

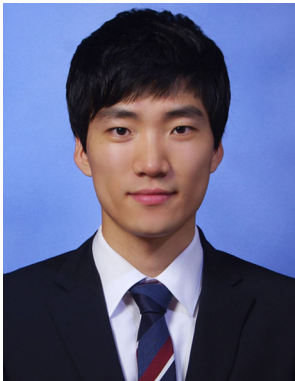
to extract strong features of the lanes. Moreover, the Feedback RANSAC algorithm is proposed to avoid false lane detection by computing two scoring functions based on the lane edge angles and the lane curvature history. Finally, the false lanes caused by obstacles are prevented by the verification using lane classification, and Kalman filter is applied for smooth lane tracking. As the results, the accuracy of 0.989 and precision of 0.999 for lane detection and tracking performance are much higher than those of existing algorithms because the proposed algorithm successfully detects lanes under aforementioned challenging scenarios. The processing time of the algorithm takes 0.677 s per frame on MATLAB, which is about 0.1 s faster than the runtime of the state-of-the-art algorithm using MATLAB [14]. The future works will focus on reducing the algorithm computation time by converting into embedded code or implementing a GPU-based system to operate in real time. Improving the proposed algorithm by considering the urban driving scenario is also remained as the future work to reduce the False Positive Rate in lane detection.

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